

$$f(x) = x + \frac{1}{x}$$

injective or not

$\mathbb{R} \setminus \{0\}$

$$x_1, x_2 \in \mathbb{R} \setminus \{0\} \quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

$$x_1 + \frac{1}{x_2} =$$

$\mathbb{Q}$

-2
2, $\frac{1}{2}$

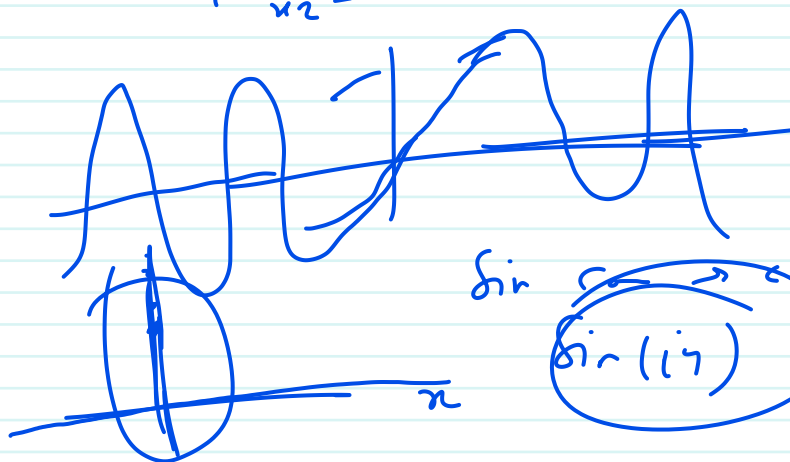
$2 + \frac{1}{2} =$

$\frac{1}{2}$

$\frac{5}{2}$

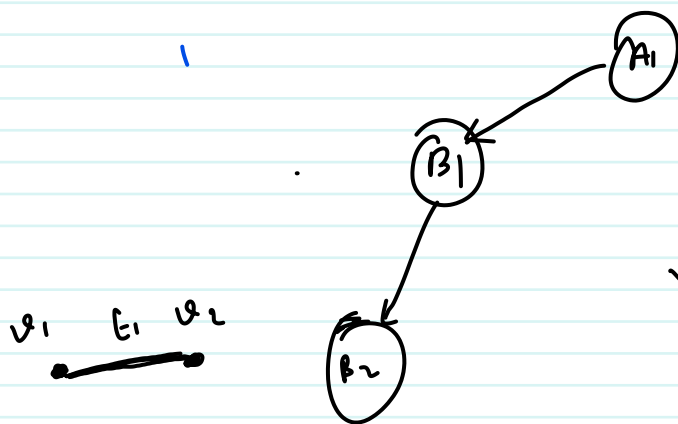
$\frac{1}{2} + 2 = \frac{5}{2}$

$x + iy$



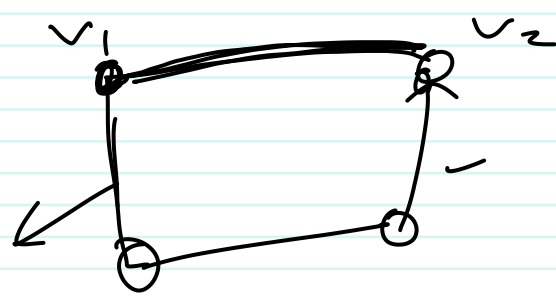
complex

$\sin \rightarrow \text{complex}$
 $\sin(iy)$



$v_1 \xrightarrow{E} v_2$
 $(\vec{v}_1, \vec{v}_2)$

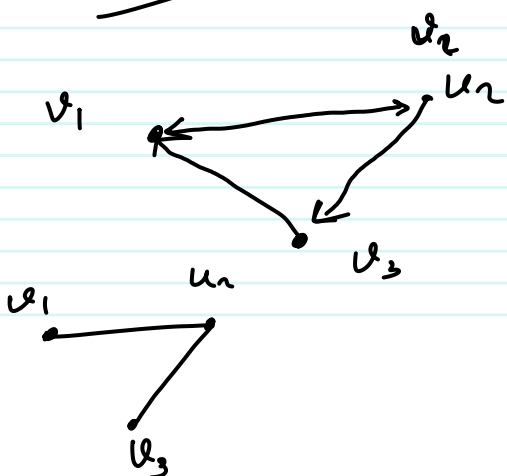
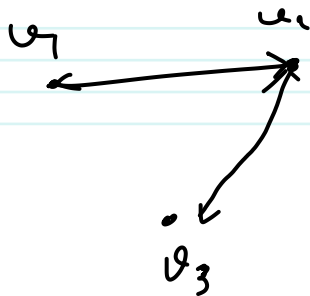
$G_1 = (V, E)$



$G_2 = (V, E)$

Graph

Directed

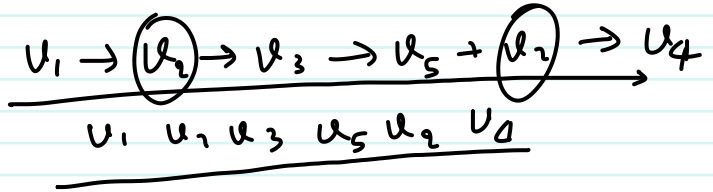
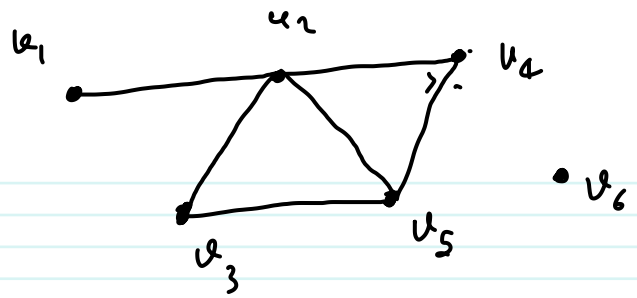


Path

Reachability

Walk

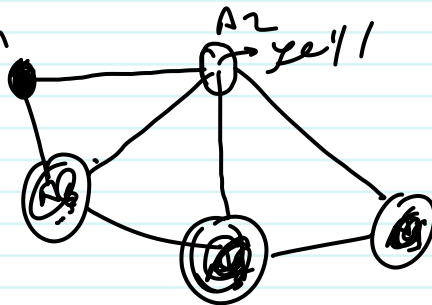
Connected



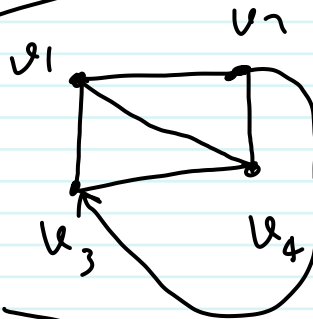
Path



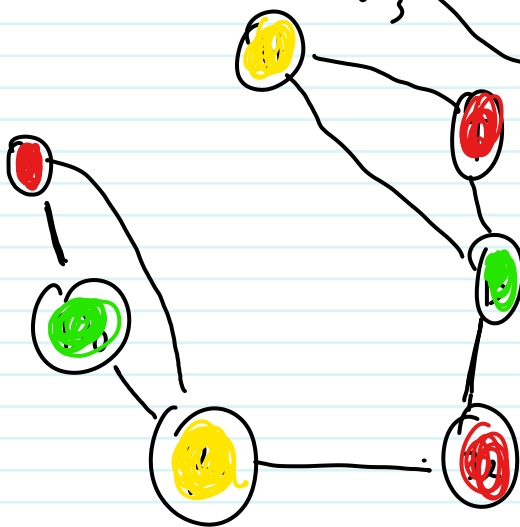
$v_1 - v_2 - \dots - v$



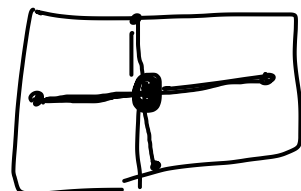
Planar

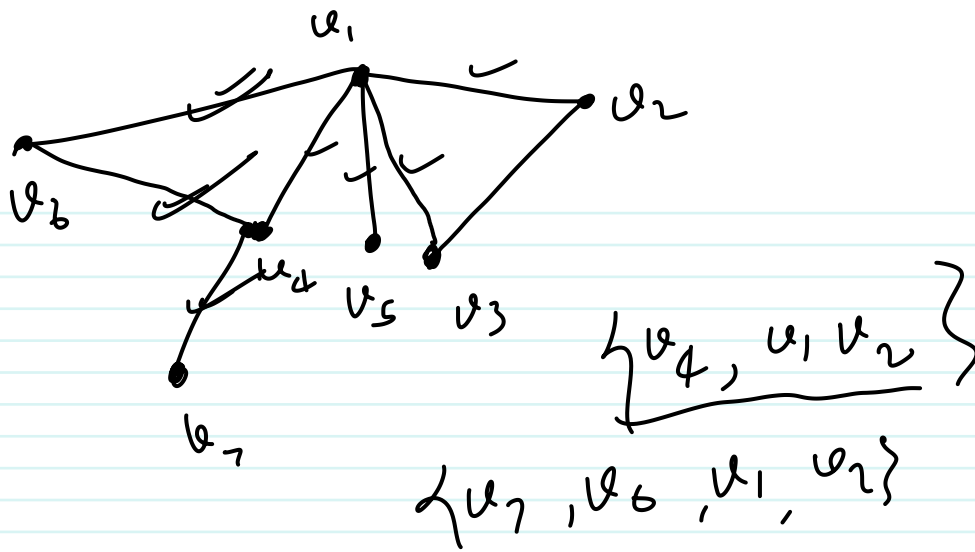


Graph
colouring

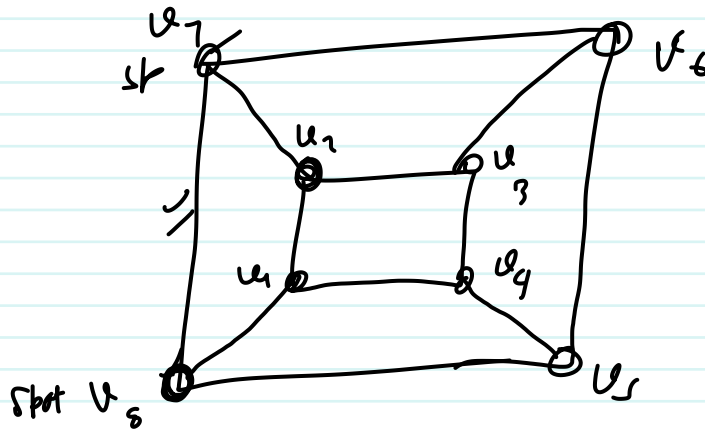


Vertex cover



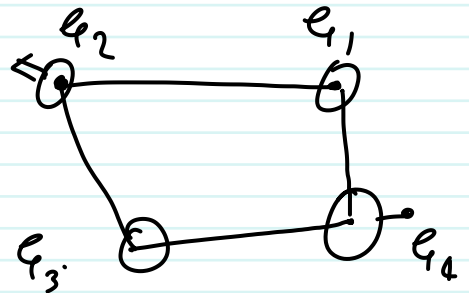


Independent set

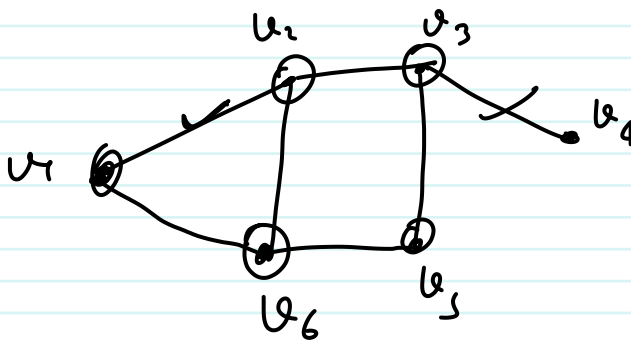


Indebn

{u5, u2, u4, u6}



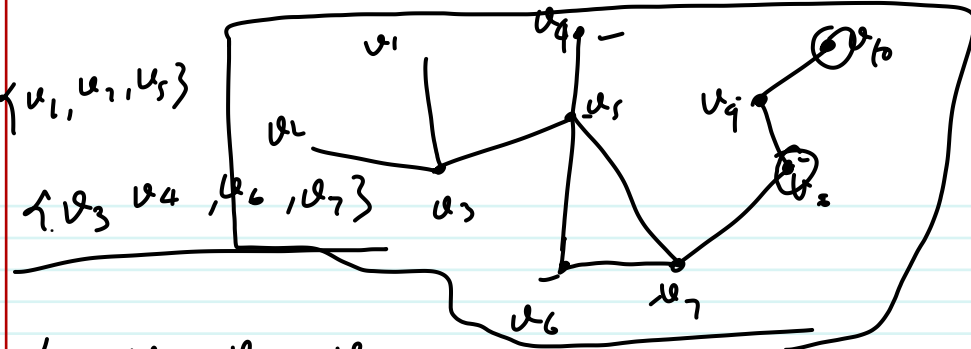
matching



{(u1, u6)}

perfect matching

u_1	$\{u_3\}$
u_2	$\{u_4\}$
u_3	
u_4	



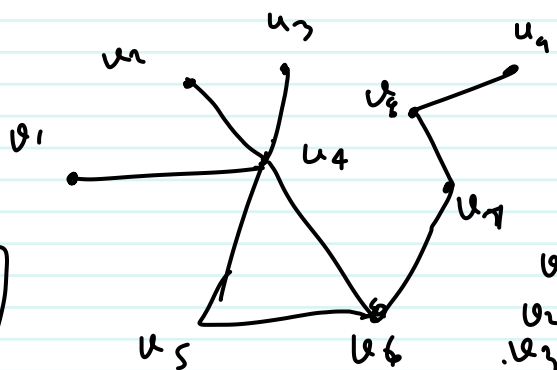
$\{u_1, u_2, u_5, u_8, u_{10}\}$

$\{u_3, u_4, u_7, u_{10}\}$

$\{u_3, u_4, u_6, u_8, u_{10}\}$

	u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	u_9	u_{10}
u_1	0	0	1	0	0	0	0	0	0	0
u_2	0	0	1	0	0	0	0	0	0	0
u_3	1	1	0	0	1	0	0	0	0	0
u_4				1	0	0	0	0	0	0
u_5					1	0	0	0	0	0
u_6						1	0	0	0	0
u_7							1	0	0	0
u_8								1	0	0
u_9									1	0
u_{10}										1

$\{u_1, u_2, u_4, u_6, u_8, u_{10}\}$



	u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	u_9
u_1	0	0	1	0	0	0	0	0	0
u_2	0	0	1	0	0	0	0	0	0
u_3	1	1	0	0	1	0	0	0	0
u_4				1	0	0	0	0	0
u_5					1	0	0	0	0
u_6						1	0	0	0
u_7							1	0	0
u_8								1	0
u_9									1

visited

u_1	T
u_2	T
u_3	T
u_4	T
u_5	T
u_6	T
u_7	T
u_8	T
u_9	T

